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Partitioning Message Passing for Graph Fraud Detection

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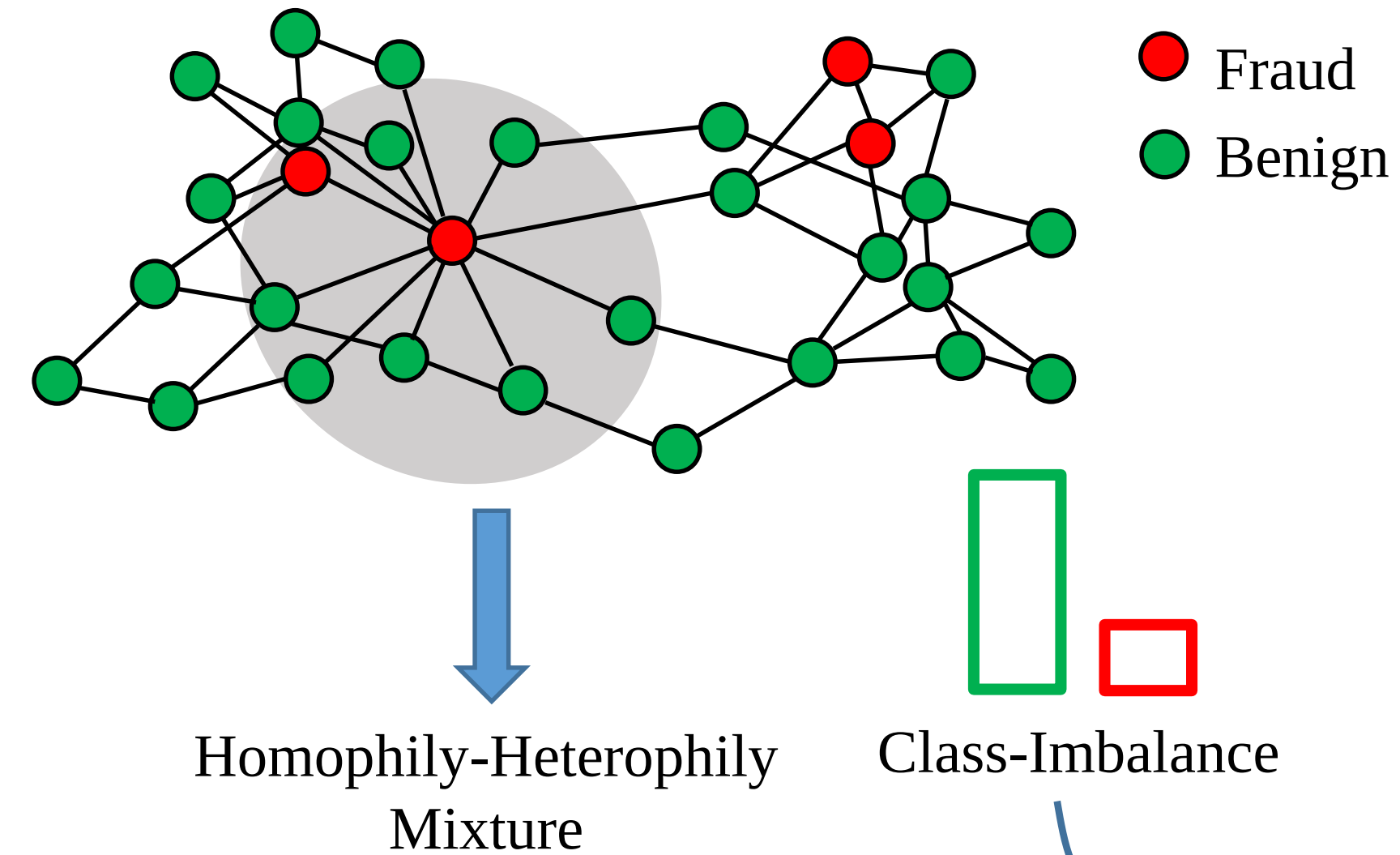
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Limitation of GNNs on Graph with Frauds

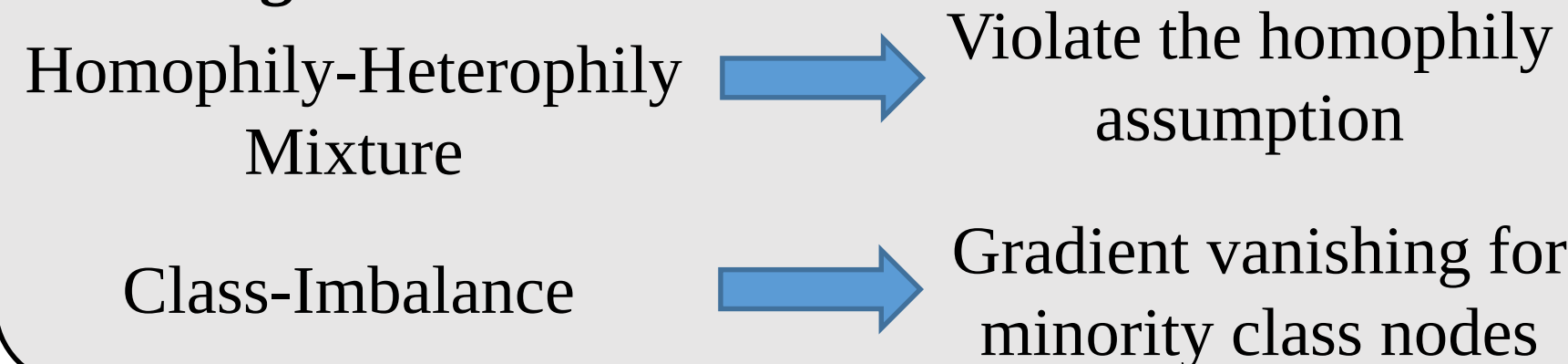


Observation:

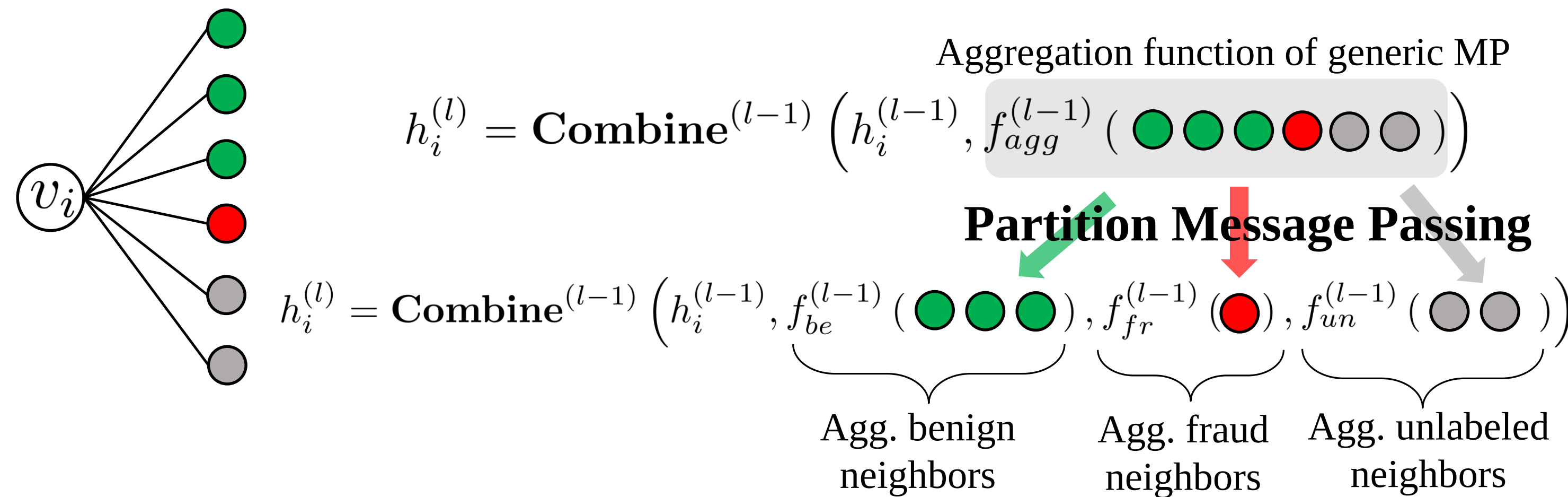
Dataset	# Nodes	# Edges	homo_ratio	Frauds (%)
YelpChi	45,954	3,846,979	0.0538	14.53%
Amazon	11,944	4,398,392	0.0512	6.87%
T-Finance	39,357	21,222,543	0.4363	4.58%
T-Social	5,781,065	73,105,508	0.1003	3.01%
Grab	~1M	~10M	-	0.54%

Network attackers often sparsely inject a small number of fraudulent nodes into the target network to camouflage their fraudulent activities and spread influence.

Challenges:



Proposed Architechure: PMP



1. Handling unlabeled neighbors as weighted combination of fraud and benign neighbors:

$$\mathbf{W}_{un}^{(l)} = \alpha_i^{(l)} \mathbf{W}_{fr}^{(l)} + (1 - \alpha_i^{(l)}) \mathbf{W}_{be}^{(l)}$$

2. Root-specific weight matrices generation:

$$\mathbf{W}_{fr,i}^{(l)} = \Psi_{fr} \left(h_i^{(l)} \right) \quad \mathbf{W}_{be,i}^{(l)} = \Psi_{be} \left(h_i^{(l)} \right)$$

Theoretical Analysis

Why is PMP effective in graphs with mixed homophily and heterophily?

Unlike most GNNs which function as global spectral graph filters, PMP operates as a node-specific adaptive filter, assigning a unique spectral filter to each node (refer to Theorem 1 of our paper) .

$$\mathbf{H}(v_i) = \mathbf{U} g_i(\Lambda) \mathbf{U}^\top \mathbf{X}$$

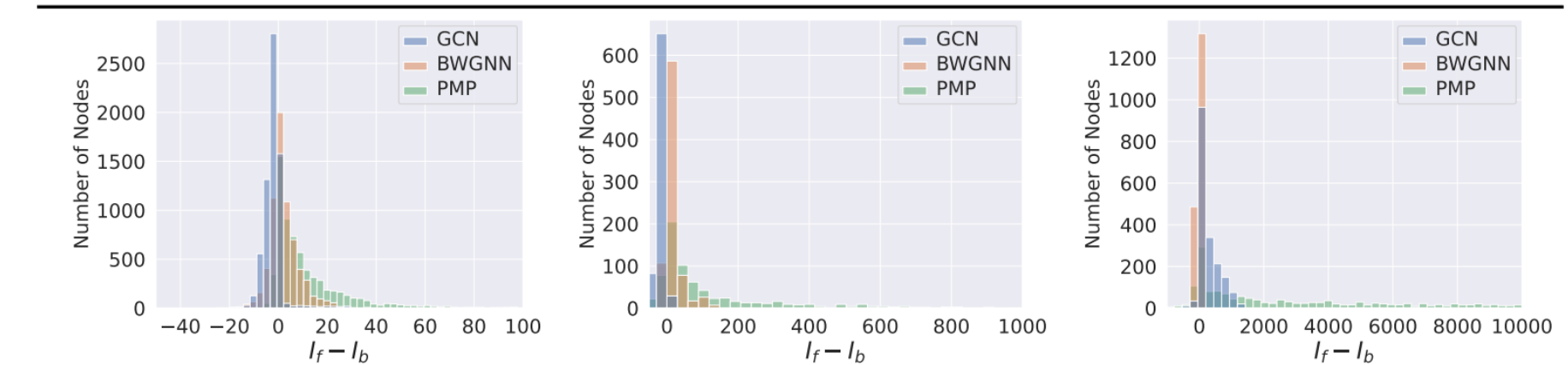
$$h_i = \mathbf{H}(v_i)_i$$

Experiments

Main results

Method	Yelp			Amazon		
	AUC	F1-Macro	G-Mean	AUC	F1-Macro	G-Mean
GCN	59.83 $\pm$ 0.49	56.20 $\pm$ 0.67	43.65 $\pm$ 2.62	83.69 $\pm$ 1.25	64.86 $\pm$ 6.94	57.18 $\pm$ 19.51
GAT	57.15 $\pm$ 0.29	48.79 $\pm$ 2.30	16.59 $\pm$ 7.89	81.02 $\pm$ 1.79	64.64 $\pm$ 3.87	66.75 $\pm$ 13.45
GraphSAGE	89.38 $\pm$ 0.19	75.46 $\pm$ 0.81	73.07 $\pm$ 4.79	93.16 $\pm$ 0.87	88.26 $\pm$ 0.62	83.46 $\pm$ 1.49
GPRGNN	82.85 $\pm$ 0.66	63.19 $\pm$ 0.80	75.62 $\pm$ 1.24	93.72 $\pm$ 0.68	80.66 $\pm$ 1.82	85.56 $\pm$ 2.73
FAGCN	74.23 $\pm$ 0.27	61.18 $\pm$ 0.75	67.37 $\pm$ 1.26	95.00 $\pm$ 0.81	87.29 $\pm$ 1.53	79.62 $\pm$ 0.96
Care-GNN	76.19 $\pm$ 2.92	63.32 $\pm$ 0.94	67.91 $\pm$ 3.59	90.67 $\pm$ 1.49	86.39 $\pm$ 1.66	70.52 $\pm$ 0.21
PC-GNN	79.87 $\pm$ 0.14	63.00 $\pm$ 2.30	71.60 $\pm$ 1.30	95.86 $\pm$ 0.14	89.56 $\pm$ 0.77	90.30 $\pm$ 0.44
H2-FDetector	89.48 $\pm$ 1.26	74.38 $\pm$ 2.42	79.15 $\pm$ 2.57	96.03 $\pm$ 0.69	86.91 $\pm$ 1.01	91.74 $\pm$ 0.47
BWGNN	90.54 $\pm$ 0.49	76.96 $\pm$ 0.89	77.12 $\pm$ 0.99	97.42 $\pm$ 0.48	91.72 $\pm$ 0.84	90.01 $\pm$ 0.36
GHRN	90.57 $\pm$ 0.36	77.54 $\pm$ 1.02	74.21 $\pm$ 1.57	97.07 $\pm$ 0.73	92.36$\pm$0.97	90.58 $\pm$ 0.45
GDN	90.34 $\pm$ 0.80	76.05 $\pm$ 0.60	80.84 $\pm$ 0.09	97.09 $\pm$ 0.16	90.68 $\pm$ 0.42	90.78 $\pm$ 0.11
PMP	93.97$\pm$0.15	81.96$\pm$0.56	83.92$\pm$1.04	97.57$\pm$0.12	92.03 $\pm$ 0.79	91.85$\pm$0.65

Method	T-Finance			T-Social		
	AUC	F1-Macro	G-Mean	AUC	F1-Macro	G-Mean
GCN	93.31 $\pm$ 0.75	86.86 $\pm$ 0.77	80.43 $\pm$ 0.82	76.00 $\pm$ 2.47	46.85 $\pm$ 1.64	71.19 $\pm$ 2.73
GAT	92.77 $\pm$ 1.12	62.19 $\pm$ 0.98	77.46 $\pm$ 2.20	63.45 $\pm$ 1.36	57.22 $\pm$ 0.81	69.53 $\pm$ 2.18
GraphSAGE	95.05 $\pm$ 0.22	90.37 $\pm$ 0.49	85.35 $\pm$ 0.38	94.19 $\pm$ 1.65	74.97 $\pm$ 2.49	72.07 $\pm$ 2.39
GPRGNN	54.82 $\pm$ 0.78	56.25 $\pm$ 0.27	32.96 $\pm$ 0.25	82.79 $\pm$ 1.21	49.23 $\pm$ 0.87	50.03 $\pm$ 7.62
FAGCN	OOM	OOM	OOM	OOM	OOM	OOM
Care-GNN	93.79 $\pm$ 0.92	82.59 $\pm$ 0.86	84.58 $\pm$ 1.26	78.91 $\pm$ 0.84	51.87 $\pm$ 1.76	63.60 $\pm$ 0.77
PC-GNN	92.09 $\pm$ 0.58	55.81 $\pm$ 0.34	81.96 $\pm$ 0.42	88.98 $\pm$ 0.66	44.13 $\pm$ 1.07	75.07 $\pm$ 0.80
H2-FDetector	94.99 $\pm$ 0.68	74.21 $\pm$ 0.70	85.91 $\pm$ 0.69	OOT	OOT	OOT
BWGNN	95.84 $\pm$ 0.46	88.66 $\pm$ 0.72	85.16 $\pm$ 1.66	94.72 $\pm$ 1.88	84.06 $\pm$ 2.89	81.51 $\pm$ 4.08
GHRN	95.77 $\pm$ 0.60	87.92 $\pm$ 0.75	81.65 $\pm$ 0.43	90.60 $\pm$ 1.78	68.28 $\pm$ 0.73	63.42 $\pm$ 2.81
GDN	95.53 $\pm$ 0.74	88.75 $\pm$ 2.26	87.84 $\pm$ 0.97	88.06 $\pm$ 0.43	56.89 $\pm$ 1.11	30.33 $\pm$ 3.67
PMP	97.10$\pm$0.23	91.90$\pm$0.50	88.51$\pm$1.26	99.62$\pm$0.07	95.27$\pm$0.32	92.97$\pm$0.64



Method	Yelp		Amazon		T-Finance	
	AUC	F1-Macro	AUC	F1-Macro	AUC	F1-Macro
GraphSAGE	89.38	75.46	93.16	88.26	95.05	90.37
+partition	93.15	78.02	96.33	89.64	95.92	91.43
++adaptive combination	93.51	79.93	97.61	91.29	96.86	91.31
+++root-specific weights (full PMP)	93.97	81.96	97.57	92.03	97.10	91.90

